Year 13 Maths - Further differentiation

Similar questions - Exponential curves and Newton-Raphson

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A curve has equation $y = x^2 e^{-x}$.

(a) Show that $\dfrac{dy}{dx} = x(2-x)e^{-x}$ and find the coordinates of the stationary points. **[4]**

(b) Show that $\dfrac{d^2y}{dx^2} = (x^2 - 4x + 2)e^{-x}$ and hence determine the nature of each stationary point. **[4]**

(c) Find the exact x-coordinates of the points of inflection. **[2]**

(d) Find the exact area of the region enclosed by the curve, the x-axis and the line $x = 2$. **[5]**

2

(a) Show that the equation $x^3 + 2x - 5 = 0$ has a root between 1 and 2. **[2]**

(b) Use the Newton–Raphson method with $x_1 = 1.2$ to find this root correct to 3 decimal places, showing each iteration. **[4]**

(c) Show that the iteration $x_{n+1} = \dfrac{5 - x_n^3}{2}$ cannot converge to this root. **[3]**

Question 1

(a) Product rule: $2xe^{-x} - x^2e^{-x} = x(2-x)e^{-x}$. Zero at $x = 0$ and $x = 2$: points $(0,0)$ and $\left(2, 4e^{-2}\right)$

(b) Differentiating again: $(2-2x)e^{-x} - (2x-x^2)e^{-x} = (x^2-4x+2)e^{-x}$. At $x = 0$ it is $2 > 0$ (minimum). At $x = 2$ it is $-2e^{-2} < 0$ (maximum)

(c) $x^2 - 4x + 2 = 0$ gives $x = 2 \pm \sqrt{2}$ (the sign of $\dfrac{d^2y}{dx^2}$ changes either side)

(d) $\displaystyle\int_0^2 x^2e^{-x}\,dx = \left[-e^{-x}(x^2+2x+2)\right]_0^2 = -10e^{-2} + 2 = 2 - 10e^{-2}$ (found by integrating by parts twice)

Question 2

(a) $f(1) = -2 < 0$ and $f(2) = 7 > 0$ and f is continuous, so there is a root in $(1,2)$

(b) $x_{n+1} = x_n - \dfrac{x_n^3 + 2x_n - 5}{3x_n^2 + 2}$: $x_2 = 1.3380$, $x_3 = 1.3283$, $x_4 = 1.3283$, $x_5 = 1.3283$. The root is 1.328 to 3 d.p.

(c) For $g(x) = \dfrac{5 - x^3}{2}$, $g'(x) = -\frac{3}{2}x^2$. Near the root $x \approx 1.33$, $|g'| \approx 2.7 > 1$, so the iteration moves away from the root and does not converge